

Owner/MC:  	ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT								Contractors:  
	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT								
Contract No: AN-C0025	Owner	Project	Phase	Unit	Discipline	Type	Serial	Rev.	Page 1 of 17
	AISPC	10	DE	930	PI	ISO	0001	A0	Class : 1B

ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT

A0	6-Jan-2026	ISSUED FOR APPROVAL	M.Imanzadeh	M. Mohammadi	F. Izadjoo
Rev.	Issue Date	Purpose of Issue	Prepared	Checked	Approved

Owner/MC:		ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT								Contractors:	
		ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT									
Contract No: AN-C0025	Owner	Project	Phase	Unit	Discipline	Type	Serial	Rev.	Page 2 of 17		
	AISPC	10	DE	930	PI	ISO	0001	A0	Class : 1B		

Revision Index

Sheet	Revisions						
	A0	A1	B0				
1	x						
2	x						
3	x						
4	x						
5	x						
6	x						
7	x						
8	x						
9	x						
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Sheet	Revisions						
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Owner/MC:						ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT										Contractors:	
 شرکت ملی پتروشیمی و انرژی ها  شرکت ملی پتروشیمی ایران																ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	
Contract No.: AN-C0025						Owner		Project	Phase	Unit	Discipline		Type	Serial	Rev.	Page 3 of 17	
						AISPC		10	DE	930	PI		ISO	0001	A0	Class : 1B	
ITEM NO.	LINE SIZE	FLUID	AREA NO.	SERIAL NO.	PIPING CLASS	INSULATION TYPE	ISOMETRIC SHEET NO.	ISOMETRIC STATUS	ISO LIST DATE	ISO LIST REV.	ISOMETRIC REVA0	ISOMETRIC REVA1	ISOMETRIC REVB0	ISOMETRIC REVB1	SECTION	Remark	
1	28"	CWR	10	930065A	DA01	UCW	1	IFA	06/01/2026	A0	06/01/2026						
2	28"	CWR	10	930067A	DA01	UCW	1	IFA	06/01/2026	A0	06/01/2026						
3	28"	CWR	10	930069A	DA01	UCW	1	IFA	06/01/2026	A0	06/01/2026						
4	28"	CWR	10	930072A	DA01	UCW	1	IFA	06/01/2026	A0	06/01/2026						
5	28"	CWR	10	930074A	DA01	UCW	1	IFA	06/01/2026	A0	06/01/2026						
6	28"	CWR	10	930076A	DA01	UCW	1	IFA	06/01/2026	A0	06/01/2026						
7	60"	CWR	10	930001A	DA01	UCW	4	IFA	06/01/2026	A0	06/01/2026						
8	60"	CWR	10	930002A	DA01	UCW	1	IFA	06/01/2026	A0	06/01/2026						
9	60"	CWR	10	930027A	DA01	UCW	1	IFA	06/01/2026	A0	06/01/2026						
10	60"	CWR	10	930030A	DA01	UCW	2	IFA	06/01/2026	A0	06/01/2026						
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CONT. ON
28"-CWR-10-930066A-DA01-N
E 837391
N 1467586
EL +295398

F2

<1>

28"NS

3336

28"NS

CONT. ON
60"-CWR-10-930002A-DA01-UCW
E 837391
N 1467586
EL +292062

BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E29OWL	3.3M
FLANGES				
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	3211	28			

NOTE

1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
6-ISOMETRIC DRAWING FOR 1½" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION		DESIGN		OPERATING						
PRESS. (barg):		7.4		2.3						
TEMP.(°C)		65		43						
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER								
INSULATION TYPE:			INSULATION THK.: -							
PAINT CODE: -			FIRE PROOFING: NO							
STRESS ANALYSIS:										
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT		A0	06/JAN/2026	ISSUED FOR REVIEW	M. IMANZADEH	M. MOHAMMADI	F. IZADJOO
			PWHT: -		REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
	ELEC. TRACING:		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)					

CLIENT:		MC:	CONTRACTOR:	
 شرکت پتروشیمی بن سینا Petro. Sh. Petrochemical Industry Co PGPIC تحت نظر بنیادی مرکزی راهبردی		 شرکت صنعت پتروشیمی بن سینا Petro. Sh. Petrochemical Industry Co PGPIC فرماندهی مناطق ویژه مناطق اقتصادی	 شرکت پتروشیمی بن سینا Petro. Sh. Petrochemical Industry Co PGPIC فرماندهی مناطق ویژه مناطق اقتصادی	
PROJECT: ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT				
LINE No.:		28"-CWR-10-930065A-DA01-UCW		REV: 1B CLASS: 1 OF 1
PROJECT DOCUMENT No.:		PROJECT DOCUMENT TITLE:		REV: AREA No.:
AISPC-10-DE-930-P1-ISO-0001		ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30		130



CONT. ON
28"-CWR-10-930068A-DA01-N
E 832409
N 1467586
EL +295398

F2

<1>

28"NS

3336

28"NS

CONT. ON
60"-CWR-10-930002A-DA01-UCW
E 832409
N 1467586
EL +292062

BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E29OWL	3.3M
FLANGES				
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	3211	28			

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TEMP.(°C)		65		43						
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER								
INSULATION TYPE:			INSULATION THK.: -							
PAINT CODE: -			FIRE PROOFING: NO							
STRESS ANALYSIS:										
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	A0	06/JAN/2026	ISSUED FOR REVIEW		M. IMANZADEH	M. MOHAMMADI	F. IZADJOO
			PWHT: -	REV.	DATE	DESCRIPTION		ISSUED BY	CHECKED BY	APPROVED BY
	ELEC. TRACING:		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)					

CLIENT:		MC:	CONTRACTOR:	
 شرکت پتروشیمی ابن سینا Parvaz Oil Petrochemical Industries Co PGPIC تحت نظر مهندس مشاور		 شرکت صنعتی پتروشیمی ابن سینا Parvaz Oil Petrochemical Industries Co PGPIC فروش خطوط لوله فلزی و غیر فلزی		 شرکت پتروشیمی ابن سینا مهندسین مشاور چگالتاش
PROJECT: ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT				
LINE No.:		28"-CWR-10-930067A-DA01-UCW		REV: 1B CLASS: 1 OF 1
PROJECT DOCUMENT No.:		PROJECT DOCUMENT TITLE:		REV: AREA No.:
AISPC-10-DE-930-P1-ISO-0001		ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30		130



CONT. ON
28"-CWR-10-930070A-DA01-N
E 806891
N 1467586
EL +295398

F2

<1>

28"NS

3336

28"NS

CONT. ON
60"-CWR-10-930002A-DA01-UCW
E 806891
N 1467586
EL +292062

BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E29OWL	3.3M
FLANGES				
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	3211	28			

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TEMP.(°C)		65		43						
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER								
INSULATION TYPE:			INSULATION THK.: -							
PAINT CODE: -			FIRE PROOFING: NO							
STRESS ANALYSIS:										
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	A0	06/JAN/2026	ISSUED FOR REVIEW		M. IMANZADEH	M. MOHAMMADI	F. IZADJOO
			PWHT: -	REV.	DATE	DESCRIPTION		ISSUED BY	CHECKED BY	APPROVED BY
	ELEC. TRACING:		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)					

CLIENT:		MC:	CONTRACTOR:	
				
PROJECT: ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT				
LINE No.:		28"-CWR-10-930069A-DA01-UCW		REV: 1B CLASS: 1 OF 1
PROJECT DOCUMENT No.:		PROJECT DOCUMENT TITLE:		REV: AREA No.:
AISPC-10-DE-930-P1-ISO-0001		ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30		130



CONT. ON
28"-CWR-10-930071A-DA01-N
E 771409
N 1467586
EL +295398

F2

<1> 28"NS

3336

28"NS

CONT. ON
60"-CWR-10-930002A-DA01-UCW
E 771409
N 1467586
EL +292062

BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E29OWL	3.3M
FLANGES				
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

CUT PIPE LENGTH

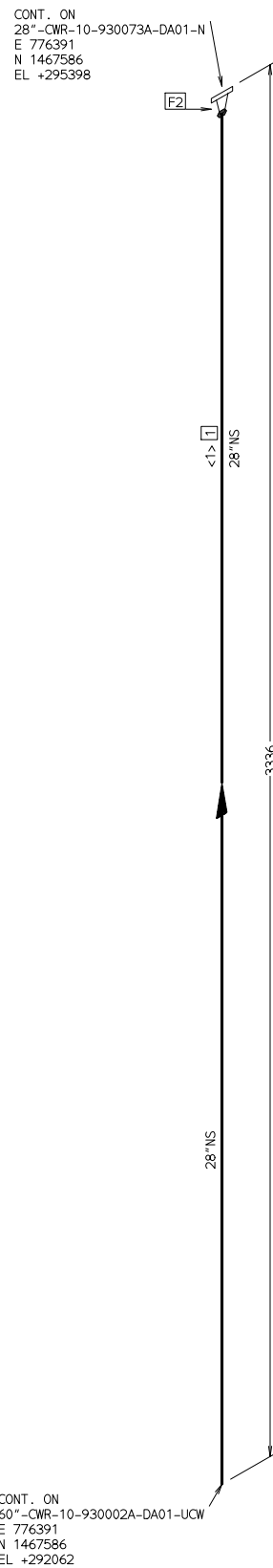
PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	3211	28			

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CONDITION		DESIGN		OPERATING						
PRESS. (barg):		7.4		2.3						
TEMP.(°C)		65		43						
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER								
INSULATION TYPE:			INSULATION THK.: -							
PAINT CODE: -			FIRE PROOFING: NO							
STRESS ANALYSIS:										
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	A0	06/JAN/2026	ISSUED FOR REVIEW		M. IMANZADEH	M. MOHAMMADI	F. IZADJOO
			PWHT: -	REV.	DATE	DESCRIPTION		ISSUED BY	CHECKED BY	APPROVED BY
	ELEC. TRACING:		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)					

CLIENT:  شرکت پتروشیمی ابن سینا Parvaz Oil Petrochemical Industries Co PGPIC تحت نظر مهندس مشاور		MC:  شرکت صنعتی پتروشیمی ابن سینا Parvaz Oil Petrochemical Industries Co PGPIC فروش خطوط لوله فلزی و غیر فلزی		CONTRACTOR:  شرکت پتروشیمی ابن سینا Parvaz Oil Petrochemical Industries Co PGPIC تحت نظر مهندس مشاور	
PROJECT: ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT					
LINE No.: 28"-CWR-10-930072A-DA01-UCW				REV: A0	CLASS: SHT. NO. 1B 1 OF 1
PROJECT DOCUMENT No.: AISPC-10-DE-930-P1-ISO-0001		PROJECT DOCUMENT TITLE: ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30		REV:	AREA No.: 130



CUT PIPE LENGTH		
PIECE NO <1>	CUT LENGTH 3211	N.S. (INS) 28

PIECE NO	CUT LENGTH	N.S. (INS)
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CLIENT:		MC:		CONTRACTOR:	
PROJECT:	ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT				
LINE No.:	28"-CWR-10-930074A-DA01-UCW			REV:	CLASS: SHT.NO.
				AO	1B 1 OF 1
PROJECT DOCUMENT No.:		PROJECT DOCUMENT TITLE:			
AISP-10-DE-930-PI-ISO-0001		ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30			

NOTE
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CONT. ON
28"-CWR-10-930075A-DA01-N
E 801909
N 1467586
EL +295398

F2

<1>

28"NS

3336

28"NS

CONT. ON
60"-CWR-10-930002A-DA01-UCW
E 801909
N 1467586
EL +292062

BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E29OWL	3.3M
FLANGES				
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

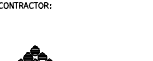
CUT PIPE LENGTH

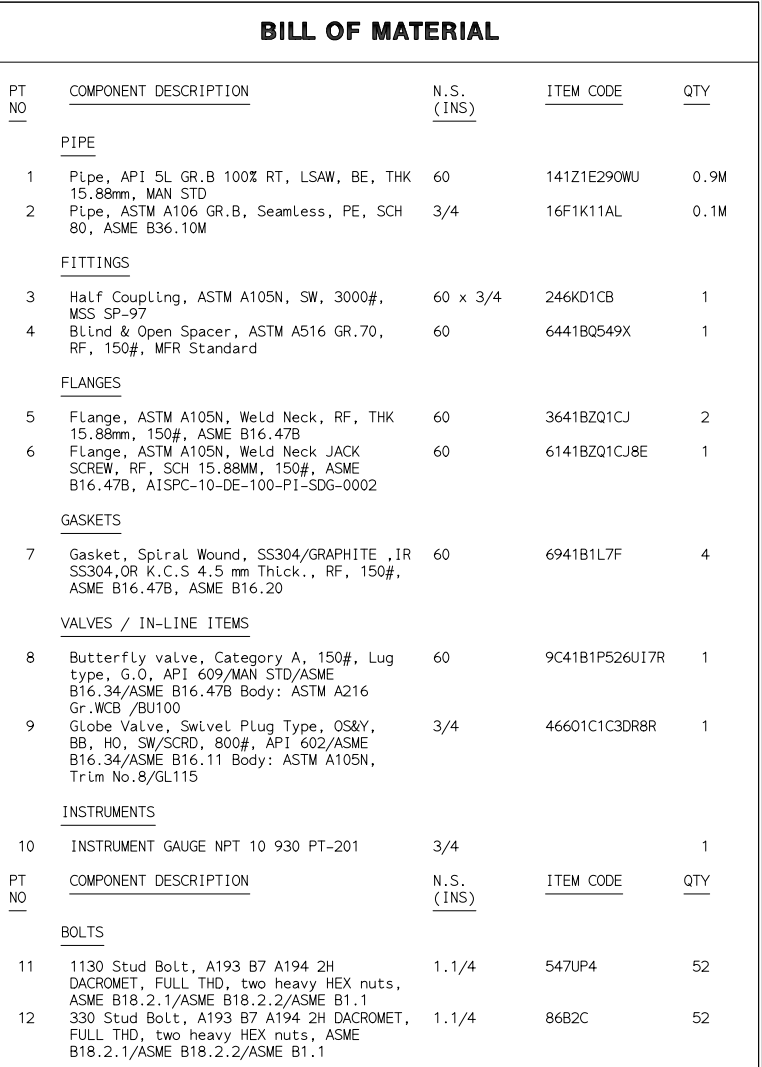
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<1>	3211	28			

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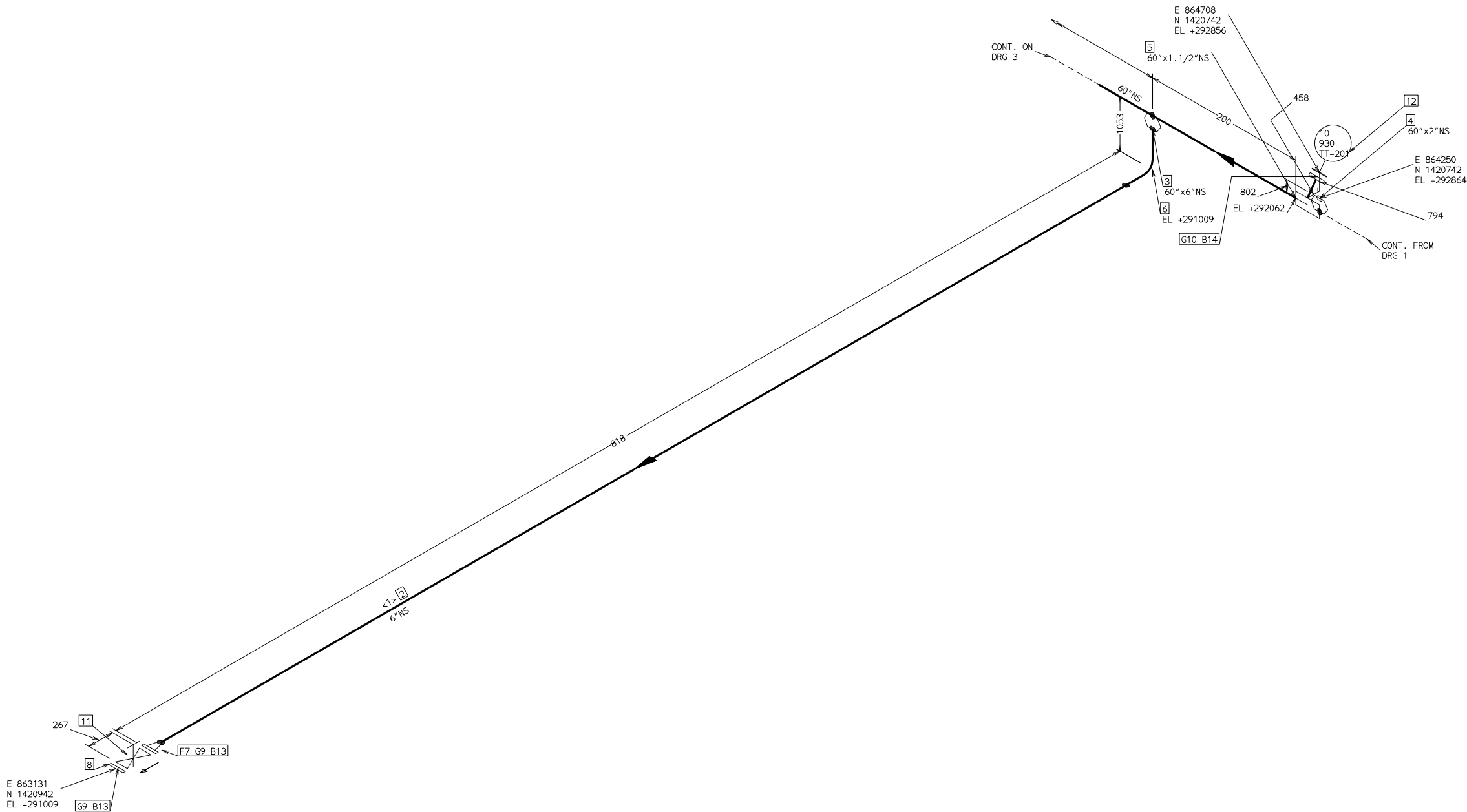
CONDITION		DESIGN		OPERATING						
PRESS. (barg):		7.4		2.3						
TEMP.(°C)		65		43						
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER								
INSULATION TYPE:			INSULATION THK.: -							
PAINT CODE: -			FIRE PROOFING: NO							
STRESS ANALYSIS:										
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	A0	06/JAN/2026	ISSUED FOR REVIEW	M. IMANZADEH	M. MOHAMMADI	F. IZADJOO	
			PWHT: -	REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY	
	ELEC. TRACING:		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)					

CLIENT:		MC:	CONTRACTOR:	
 شرکت پتروشیمی اندیمشک فارس Persian Oil Petrochemical Industries Co PGPIC گت پتروشیمی اندیمشک فارس		 شرکت صنعت پتروشیمی اندیمشک فارس Persian Oil Petrochemical Industries Co PGPIC فوق بدون توجه حاج ابراهیمی مدنی		 شرکت پتروشیمی اندیمشک فارس Persian Oil Petrochemical Industries Co PGPIC شرکت پتروشیمی اندیمشک فارس
PROJECT:				
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT				
LINE No.:		28"-CWR-10-930076A-DA01-UCW		REV: 1A0 CLASS: 1B SHT. NO. 1 OF 1
PROJECT DOCUMENT No.:		PROJECT DOCUMENT TITLE:		REV: AREA No.:
AISPC-10-DE-930-P1-ISO-0001		ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30		130



CUT PIPE LENGTH			CUT PIPE LENGTH		
PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	600	60	<2>	100	3/4
<3>	300	60			

CLIENT: 		MC: 		CONTRACTOR: 			
PROJECT: ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT							
LINE No.: 60"-CWR-10-930001A-DA01-UCW		REV/AO 1B		CLASS: 1 OF 4		SHT. NO. 1	
PROJECT DOCUMENT No.: AISPC-10-DE-930-PI-IS0-0001		PROJECT DOCUMENT TITLE: ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT		REV: AREA No.: 30			



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	0.3M
2	Pipe, ASTM A106 GR.B, SeamLess, BE, SCH 40, ASME B36.10M	6	114D1E11AL	0.5M
FITTINGS				
3	WeldoLet, ASTM A105N, BW, THK 15.88mmxSCH 40, MSS SP-97	60 x 6	254114ZD4M1CU	1
4	WeldoLet, ASTM A105N, BW, THK 15.88mmxSCH 40, MSS SP-97 ASTM A105N	60 x 2	254110ZD4M1CU	1
5	Flange, Long Weld Neck , RF, BW, SCH 80, 300#, ASTM A105N, ASME B16.5	60 x 1.1/2	41419DFQ1CG	1
6	Elbow 90, ASTM A234 WPB, SeamLess, LR, BW, SCH 40, ASME B16.9	6	214D4M53AA	1
FLANGES				
7	Flange, ASTM A105N, Weld Neck, RF, SCH 40, 150#, ASME B16.5	6	3614BDQ1CG	1
8	Flange, BLind , ASTM A105N, RF, 150#, ASME B16.5	6	3814BQ1CG	1
GASKETS				
9	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thick., RF, 150#, ASME B16.5, ASME B16.20	6	6914B1L2k	2
10	Gasket, SS304/GRAPHITE, IR SS304, OR K.C.S, Spiral Wound, 4.5 mm Thick., RF, 300#, ASME B16.5, ASME B16.20	1.1/2	699D1L2k	1
VALVES / IN-LINE ITEMS				
11	Gate Valve, Flexible Wedge, BB, OS&Y, HO, FLGD, RF, 150#, API 600/ASME B16.34/ASME B16.5 Body: ASTM A216 Gr.WCB, Trim No.8/GA101	6	4814BQ526PR8R	1
INSTRUMENTS				
12	INSTRUMENT GAUGE #300 RF 10 930 TT-201	1.1/2		1
PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
BOLTS				
13	110 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BK	16
14	100 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BI	4

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	500	6			

NOTE

1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
6-ISOMETRIC DRAWING FOR 1½" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION		DESIGN		OPERATING							
PRESS. (barg):		7.4		2.3							
TEMP.(°C)		65		43							
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER									
INSULATION TYPE:			INSULATION THK.: -								
PAINT CODE: -			FIRE PROOFING: NO								
STRESS ANALYSIS:											
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT		A0	06/JAN/2026	ISSUED FOR REVIEW		M. IMANZADEH	M. MOHAMMADI	F. IZADJOO
			PWHT: -		REV.	DATE	DESCRIPTION		ISSUED BY	CHECKED BY	APPROVED BY
	ELEC. TRACING: NO		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)						

CLIENT:

MC:

CONTRACTOR:

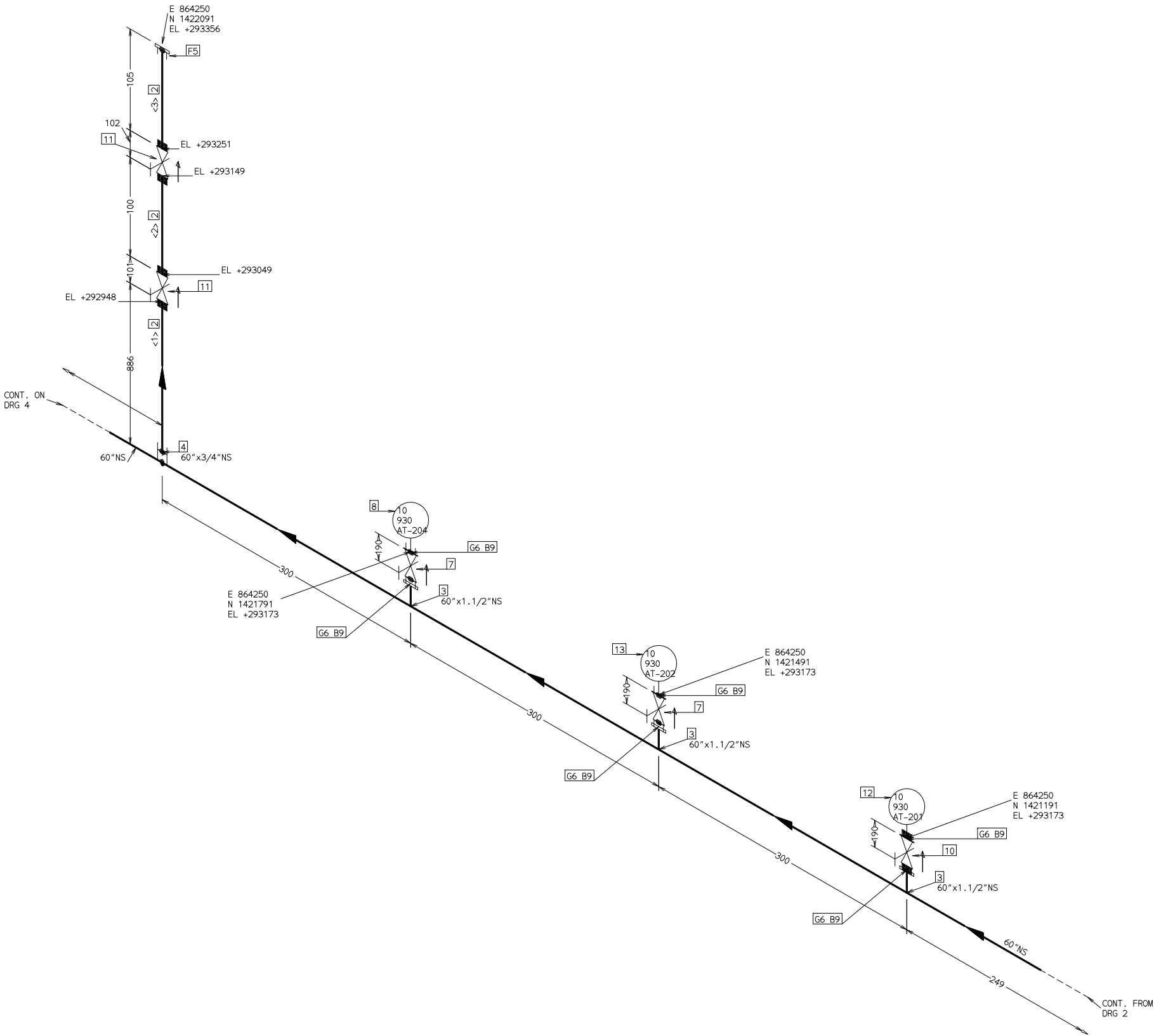
PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:
60"-CWR-10-930001A-DA01-UCW

PROJECT DOCUMENT No.:
AISPC-10-DE-930-P1-ISO-0001

PROJECT DOCUMENT TITLE:
ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30

REV: 1B
CLASS: 1B
SHT.NO: 2 OF 4
AREA No.: 130



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	1.2M
2	Pipe, ASTM A106 GR.B, SeamLess, PE, SCH 80, ASME B36.10M	3/4	16F1K11AL	0.3M
FITTINGS				
3	Flange, Long Weld Neck , RF, BW, SCH 80, 300#, ASTM A105N, ASME B16.5	60 x 1.1/2	41419DFQ1CG	3
4	Half Coupling, ASTM A105N, SW, 3000#, MSS SP-97	60 x 3/4	246KD1CB	1
FLANGES				
5	Flange, ASTM A105N, Socket welded, RF, SCH80, 150#, ASME B16.5	3/4	356BFQ1CG	1
GASKETS				
6	Gasket, SS304/GRAPHITE, IR SS304, OR K.C.S, Spiral Wound, 4.5 mm Thick., RF, 300#, ASME B16.5, ASME B16.20	1.1/2	699D1L2k	6
VALVES / IN-LINE ITEMS				
7	Gate Valve, Solid Wedge, BB, OS&Y, HO, FLGD, RF, B16.5, 300#, API 600 Body: ASTM A216 Gr.WCB, Trim No.8/GA101	1.1/2	489BQ1C8PR8R	2
INSTRUMENTS				
8	INSTRUMENT GAUGE #300 RF 10 930 AT-204	1.1/2		1
PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
BOLTS				
9	100 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BI	24
VALVES / IN-LINE ITEMS				
10	Gate Valve, Solid Wedge, BB, OS&Y, HO, FLGD, RF, B16.5, 300#, API 600 Body: ASTM A216 Gr.WCB, Trim No.8/GA101	1.1/2	489BQ1C8PR8R	1
11	Gate Valve, Solid Wedge, BB, OS&Y, HO, SW, 800#, API 602 Body: ASTM A105N, Trim No.8/GA122	3/4	486OD1C3DR8R	2
INSTRUMENTS				
12	INSTRUMENT GAUGE #300 RF 10 930 AT-201	1.1/2		1
13	INSTRUMENT GAUGE #300 RF 10 930 AT-202	1.1/2		1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	448	60	<2>	299	60
<3>	299	60	<4>	100	3/4
<5>	100	3/4	<6>	100	3/4

NOTE

- 1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
- 2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
- 3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
- 4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
- 5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
- 6-ISOMETRIC DRAWING FOR 1½" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
- 7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION		DESIGN		OPERATING							
PRESS. (barg):		7.4		2.3							
TEMP.(°C)		65		43							
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER									
INSULATION TYPE:			INSULATION THK.: -								
PAINT CODE: -			FIRE PROOFING: NO								
STRESS ANALYSIS:											
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	A0	06/JAN/2026	ISSUED FOR REVIEW		M. IMANZADEH	M. MOHAMMADI	F. IZADJOO	
			PWHT: -	REV.	DATE	DESCRIPTION		ISSUED BY	CHECKED BY	APPROVED BY	
	ELEC. TRACING: NO		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)						

CLIENT:

PGPIC

MC:

PGPIC

CONTRACTOR:

ANDIMESHK

PROJECT:

ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:60"-CWR-10-930001A-DA01-UCW

PROJECT DOCUMENT No.:AISPC-10-DE-930-P1-ISO-0001

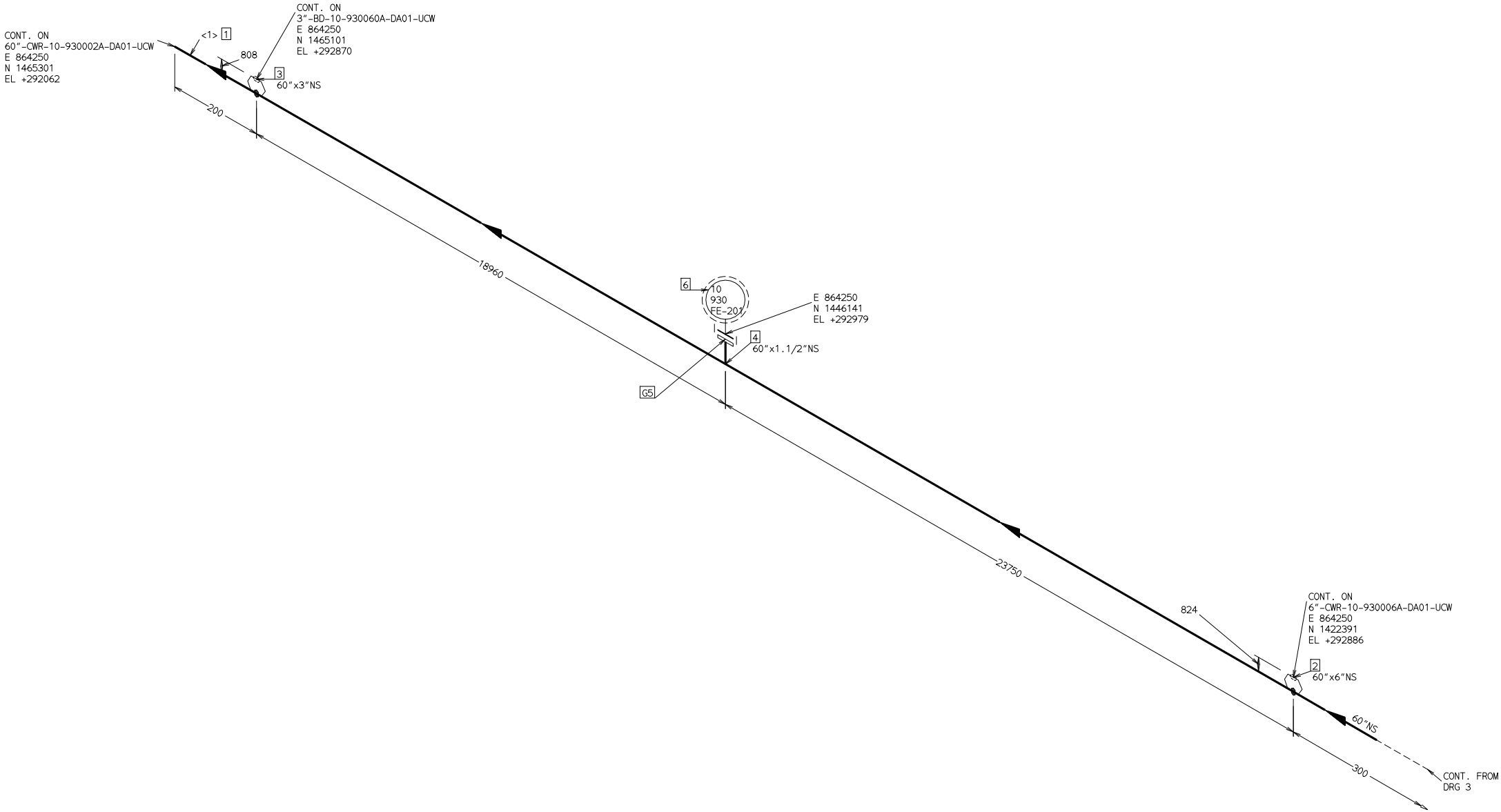
PROJECT DOCUMENT TITLE:ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30

REV:1B

CLASS:1B

SHT.NO.:3 OF 4

AREA No.:130



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	43.3M
FITTINGS				
2	WeldoLet, ASTM A105N, BW, THK 15.88mmxSCH 40, MSS SP-97	60 x 6	254114ZD4M1CU	1
3	WeldoLet, ASTM A105N, BW, THK 15.88mmxSCH 40, MSS SP-97 ASTM A105N	60 x 3	254112ZD4M1CU	1
4	Flange, Long Weld Neck , RF, BW, SCH 80, 300#, ASTM A105N, ASME B16.5	60 x 1.1/2	41419DFQ1CG	1
GASKETS				
5	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thck., RF, 150#, ASME B16.5, ASME B16.20	1.1/2	699B1L2k	1
INSTRUMENTS				
6	INSTRUMENT GAUGE #150 RF 10 930 FE-201	1.1/2		1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	24349	60	<2>	19160	60

NOTE

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7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION		DESIGN		OPERATING					
PRESS. (barg):		7.4		2.3					
TEMP.(°C)		65		43					
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER							
INSULATION TYPE:		INSULATION THK.: -							
PAINT CODE: -		FIRE PROOFING: NO							
STRESS ANALYSIS:									
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	A0	06/JAN/2026	ISSUED FOR REVIEW	M. IMANZADEH	M. MOHAMMADI	F. IZADJOO
			PWHT: -	REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
	ELEC. TRACING: NO		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)				

CLIENT:

MC:

CONTRACTOR:

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

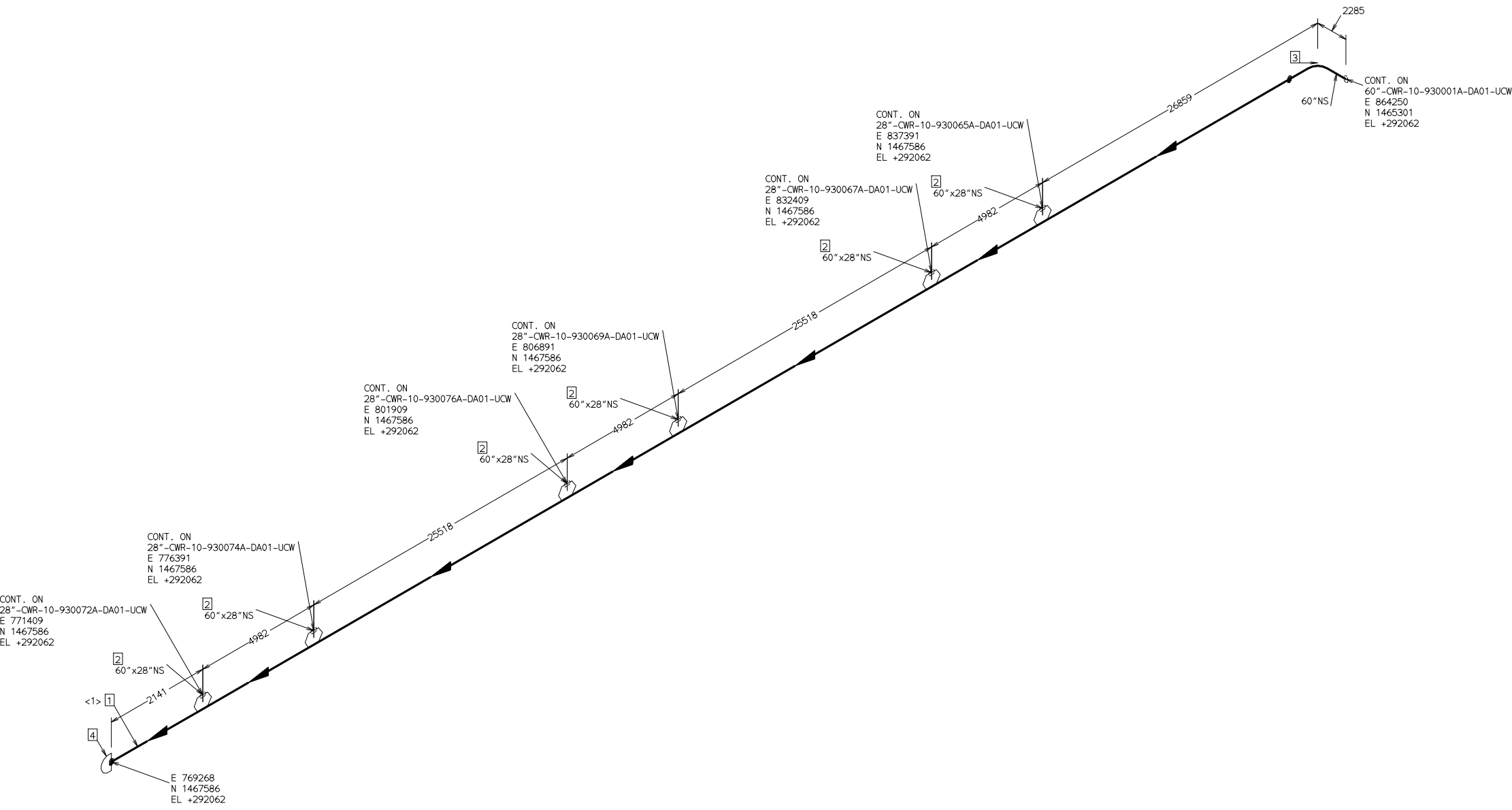
LINE No.:
60"-CWR-10-930001A-DA01-UCW

REV: 1B
CLASS: 1B
SHT.NO: 4 OF 4

PROJECT DOCUMENT No.:
AISPC-10-DE-930-P1-ISO-0001

PROJECT DOCUMENT TITLE:
ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30

REV: 1B
AREA No.: 30



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	92.7M
FITTINGS				
2	PIPE TO PIPE	60 x 28	-	6
3	Elbow 90, ASTM A234 WPB, Welded, LR, BW, THK 15.88mm, ASME B16.9/MAN STD	60	241Z4M530W2Q	1
4	Cap, ASTM A234 WPB, Welded, BW, SCH 15.88MM, ASME B16.9	60	1741Z4M53A2Q	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	92697	60			

NOTE

- 1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
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- 3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
- 4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
- 5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
- 6-ISOMETRIC DRAWING FOR 1½" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
- 7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION		DESIGN		OPERATING						
PRESS. (barg):		7.4		2.3						
TEMP.(°C)		65		43						
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER								
INSULATION TYPE:			INSULATION THK.: -							
PAINT CODE: -			FIRE PROOFING: NO							
STRESS ANALYSIS:										
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT		A0	06/JAN/2026	ISSUED FOR REVIEW	M. IMANZADEH	M. MOHAMMADI	F. IZADJOO
			PWHT: -		REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
	ELEC. TRACING: NO		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)					

CLIENT:

MC:

CONTRACTOR:

PROJECT:ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:60"-CWR-10-930002A-DA01-UCW

PROJECT DOCUMENT No.:AISPC-10-DE-930-P1-ISO-0001

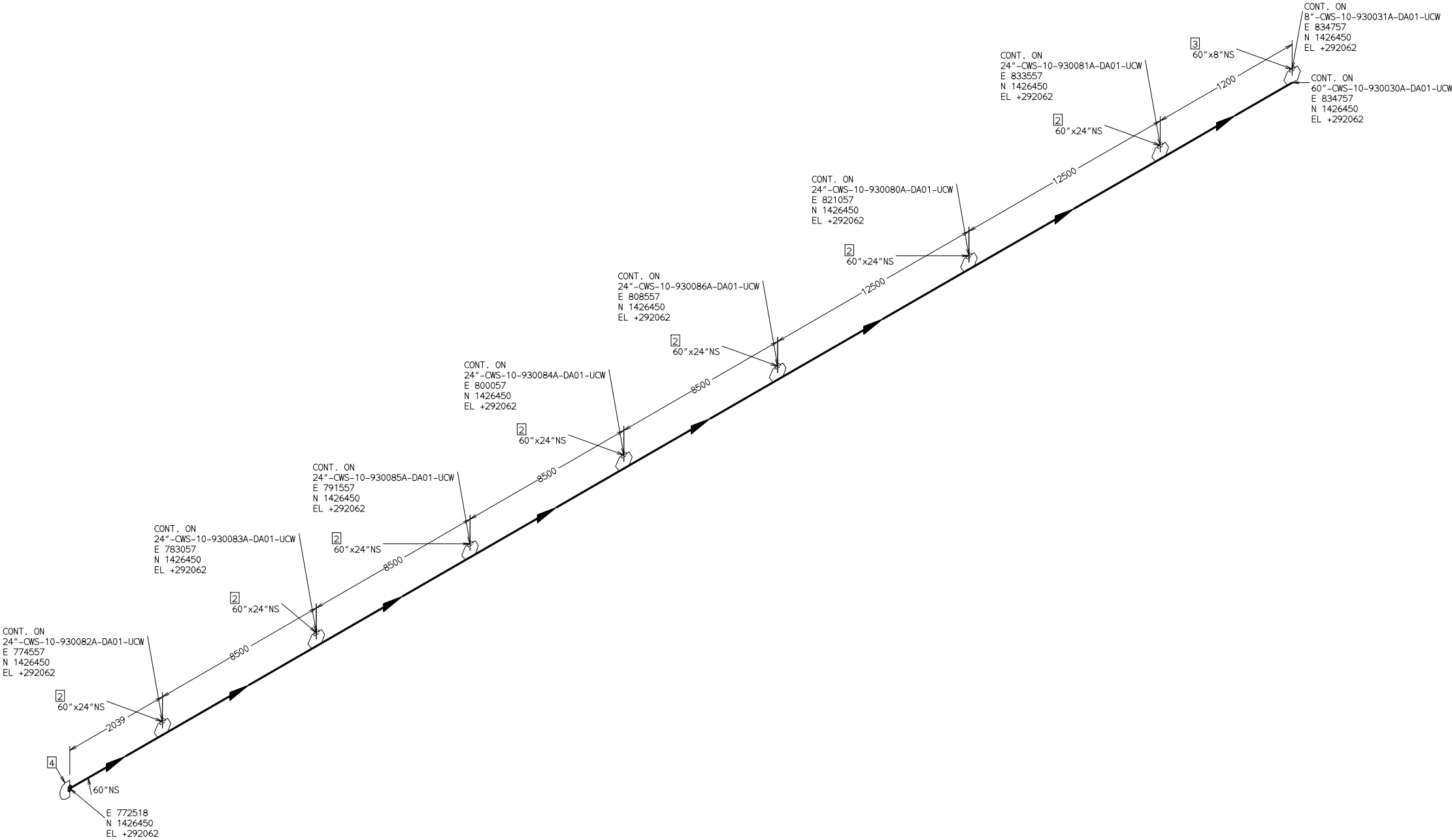
PROJECT DOCUMENT TITLE:ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30

REV: 1B

CLASS: 1B

SHT. NO.: 1 OF 1

AREA No.: 130



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	PIPE, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	62.3M
FITTINGS				
2	PIPE TO PIPE	60 x 24		7
3	PIPE TO PIPE	60 x 8		1
4	Cap, ASTM A234 WPB, Welded, BW, SCH 15.88MM, ASME B16.9	60	1741Z4M53A2Q	1

CUT PIPE LENGTH				
PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH
<1>	62239	60		

NOTE

- 1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
- 2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
- 3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
- 4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
- 5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
- 6-ISOMETRIC DRAWING FOR 1½" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
- 7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION		DESIGN		OPERATING						
PRESS. (barg):		7.4		5.9						
TEMP.(°C)		65		33						
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER								
INSULATION TYPE:			INSULATION THK.: -							
PAINT CODE: -			FIRE PROOFING: NO							
STRESS ANALYSIS:										
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	A0	06/JAN/2026	ISSUED FOR REVIEW		M. IMANZADEH	M. MOHAMMADI	F. IZADJOO
			PWHT: N	REV.	DATE	DESCRIPTION		ISSUED BY	CHECKED BY	APPROVED BY
	ELEC. TRACING: NO		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)					

CLIENT:

MC:

CONTRACTOR:

PROJECT:ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:60"-CWS-10-930027A-DA01-UCW

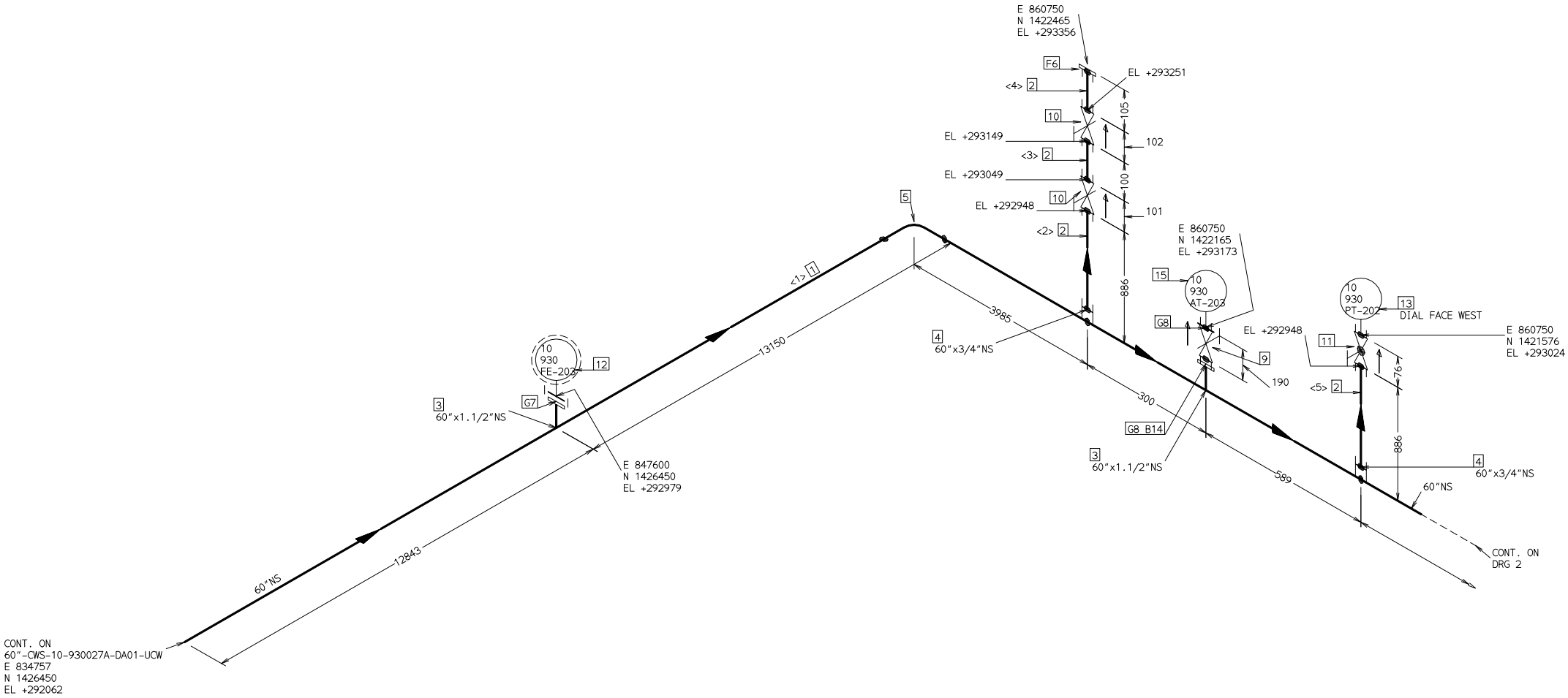
PROJECT DOCUMENT No.:AISPC-10-DE-930-P1-ISO-0001

PROJECT DOCUMENT TITLE:ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30

CLASS:1B

SHT.NO.:1 OF 1

AREA No.:1



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	26.3M
2	Pipe, ASTM A106 GR.B, SeamLess, PE, SCH 80, ASME B36.10M	3/4	16F1K11AL	0.4M
FITTINGS				
3	Flange, Long Weld Neck , RF, BW, SCH 80, 300#, ASTM A105N, ASME B16.5	60 x 1.1/2	41419DFQ1CG	2
4	Half Coupling, ASTM A105N, SW, 3000#, MSS SP-97	60 x 3/4	246KD1CB	2
5	Elbow 90, ASTM A234 WPB, Welded, LR, BW, THK 15.88mm, ASME B16.9/MAN STD	60	241Z4M53OW2Q	1
FLANGES				
6	Flange, ASTM A105N, Socket welded, RF, SCH80, 150#, ASME B16.5	3/4	356BFQ1CG	1
GASKETS				
7	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thick., RF, 150#, ASME B16.5, ASME B16.20	1.1/2	699B1L2k	1
8	Gasket, SS304/GRAPHITE, IR SS304, OR K.C.S, Spiral Wound, 4.5 mm Thick., RF, 300#, ASME B16.5, ASME B16.20	1.1/2	699D1L2k	2
VALVES / IN-LINE ITEMS				
9	Gate Valve, Solid Wedge, BB, OS&Y, HO, FLGD, RF, B16.5, 300#, API 600 Body: ASTM A216 Gr.WCB, Trim No.8/GA101	1.1/2	489BQ1C8PR8R	1
10	Gate Valve, Solid Wedge, BB, OS&Y, HO, SW, 800#, API 602 Body: ASTM A105N, Trim No.8/GA122	3/4	486OD1C3DR8R	2
11	Globe Valve, Swivel Plug Type, OS&Y, BB, HO, SW/SCRD, 800#, API 602/ASME B16.34/ASME B16.11 Body: ASTM A105N, Trim No.8/GL115	3/4	466O1C1C3DR8R	1
INSTRUMENTS				
12	INSTRUMENT GAUGE #150 RF 10 930 FE-203	1.1/2		1
13	INSTRUMENT GAUGE NPT 10 930 PT-202	3/4		1
PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
BOLTS				
14	100 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BI	4
INSTRUMENTS				
15	INSTRUMENT GAUGE #150 RF 10 930 AT-203	1.1/2		1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	12842	60	<2>	10865	60
<3>	100	3/4	<4>	100	3/4
<5>	100	3/4	<6>	1999	60
<7>	100	3/4			

NOTE

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7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION		DESIGN		OPERATING					
PRESS. (barg):		7.4		5.9					
TEMP.(°C)		65		33					
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER							
INSULATION TYPE:		INSULATION THK.: -							
PAINT CODE: -		FIRE PROOFING: NO							
STRESS ANALYSIS:									
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	A0	06/JAN/2026	ISSUED FOR REVIEW	M. IMANZADEH	M. MOHAMMADI	F. IZADJOO
			PWHT: N	REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
	ELEC. TRACING: NO		PHASE: L		REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)				

CLIENT:

شركة صنایع پتروشیمی ایران

MC:

شركة صنایع پتروشیمی ایران

CONTRACTOR:

شركة صنایع پتروشیمی ایران

PROJECT:ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:60"-CWS-10-930030A-DA01-UCW

PROJECT DOCUMENT No.:AISPC-10-DE-930-P1-ISO-0001

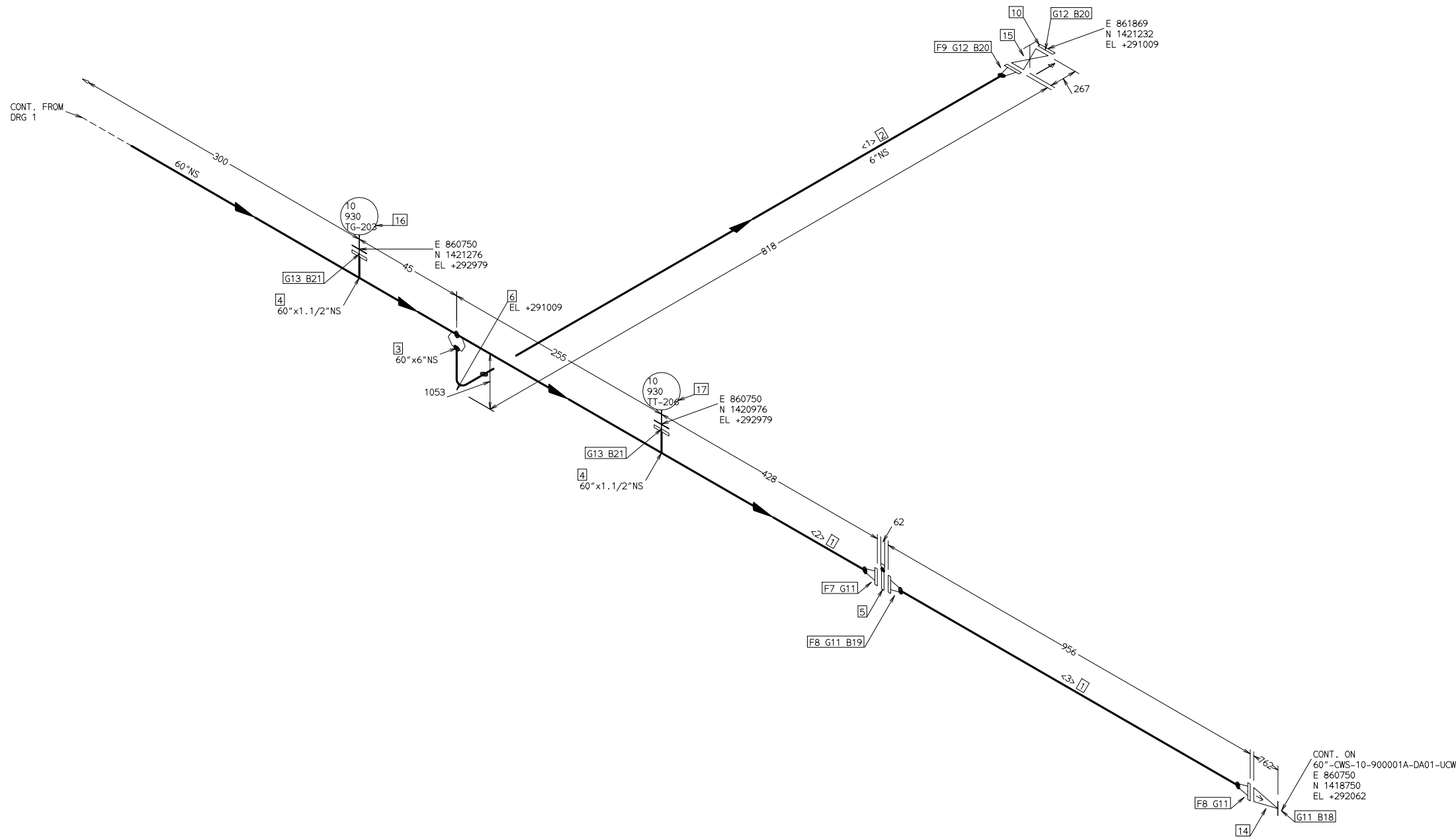
PROJECT DOCUMENT TITLE:ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30

REV:1B

CLASS:1B

SHT.NO.:1 OF 2

AREA No.:130



NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
PIPE				
1	Plpe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WJ	1.5M
2	Plpe, ASTM A106 GR.B, Seamless, BE, SCH 40, ASME B36.10M	6	114D1E11AL	0.5M
FITTINGS				
3	Weldolet, ASTM A105N, BW, THK 15.88mmxSCH 40, MSS SP-97	60 x 6	254114ZD4M1CU	1
4	Flange, Long Weld Neck , RF, BW, SCH 80, 300#, ASTM A105N, ASME B16.5	60 x 1.1/2	41419DFQ1CG	2
5	Blind & Open Spacer, ASTM A516 GR.70, RF, 150#, MFR Standard	60	6441BQ549X	1
6	Elbow 90, ASTM A234 WPB, Seamless, LR, BW, SCH 40, ASME B16.9	6	214D4M53AA	1
FLANGES				
7	Flange, ASTM A105N, Weld Neck JACK SCREW, RF, SCH 15.88MM, 150#, ASME B16.47B, A1SPC-10-DE-100-P1-SDG-0002	60	6141BZQ1CJ8E	1
8	Flange, ASTM A105N, Weld Neck, RF, THK 15.88mm, 150#, ASME B16.47B	60	3641BZQ1CJ	2
9	Flange, ASTM A105N, Weld Neck, RF, SCH 40, 150#, ASME B16.5	6	3614BDQ1CG	1
	Flange, Blind , ASTM A105N, RF, 150#, ASME B16.5	6	3814BQ1CG	1
GASKETS				
11	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thick., RF, 150#, ASME B16.47B, ASME B16.20	60	6941B1L7F	4
12	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thick., RF, 150#, ASME B16.5, ASME B16.20	6	6914B1L2k	2
13	Gasket, SS304/GRAPHITE, IR SS304, OR K.C.S, Spiral Wound, 4.5 mm Thick., RF, 300#, ASME B16.5, ASME B16.20	1.1/2	699D1L2k	2
VALVES / IN-LINE ITEMS				
14	Butterfly valve, Category A, 150#, Lug type, G.O, API 609/MAN STD/ASME B16.34/ASME B16.47B Body: ASTM A216 Gr.WCB /BU100	60	9C41B1P526U17R	1
15	Gate Valve, Flexible Wedge, BB, OS&Y, HO, FLGD, RF, 150#, API 600/ASME B16.34/ASME B16.5 Body: ASTM A216 Gr.WCB, Trim No.8/GA101	6	4814BQ526PR8R	1
INSTRUMENTS				
16	INSTRUMENT GAUGE #300 RF 10 930 TG-203	1.1/2		1
17	INSTRUMENT GAUGE #300 RF 10 930 TT-206	1.1/2		1

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	<u>BOLTS</u>			
18	1130 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	1.1/4	547UP4	52
19	330 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	1.1/4	86B2C	52
20	110 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BK	16
21	100 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BI	8

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	888	60	<2>	500	6
<3>	299	60	<4>	250	60
<5>	600	60			

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PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER							
INSULATION TYPE:		INSULATION THK.: -							
PAINT CODE: -		FIRE PROOFING: NO							
STRESS ANALYSIS:									
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	A0	06/JAN/2026	ISSUED FOR REVIEW	M. IMANZADEH	M. MOHAMMADI	F. IZADJOO
			PWHT: N	REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
	ELEC. TRACING: NO		PHASE: L		REF. P&ID DWG. No.: A1SPC-10-DE-930-PR-PID-0001-(1 of 11)				

CLIENT: 	MC: 	CONTRACTOR: 	
شرکت صنایع پتروشیمی خراسان Parsian Petrochemical Industries Co PGPIC تولید کننده انواع پلیمرها و مشتقات شرکت صنایع پتروشیمی خراسان Parsian Petrochemical Industries Co PGPIC تولید کننده انواع پلیمرها و مشتقات شرکت پترو سازه مبین مهندسین مشاور چگالش			
PROJECT: ANDIMESHK IBN SINA PETROCHEMICAL COMPANY ETHYLENE OXIDE PLANT			
LINE No.:	60"-CWS-10-930030A-DA01-UCW		
PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV: (A)	CLASS: SHT. NO. 1B 2 OF 2
AISPC-10-DE-930-P1-ISI-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT 30		